Computer Communication Networks

Function of OSI Model

Session 3

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Functions of OSI Model

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Explain Functions of:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer



Functions of OSI Model

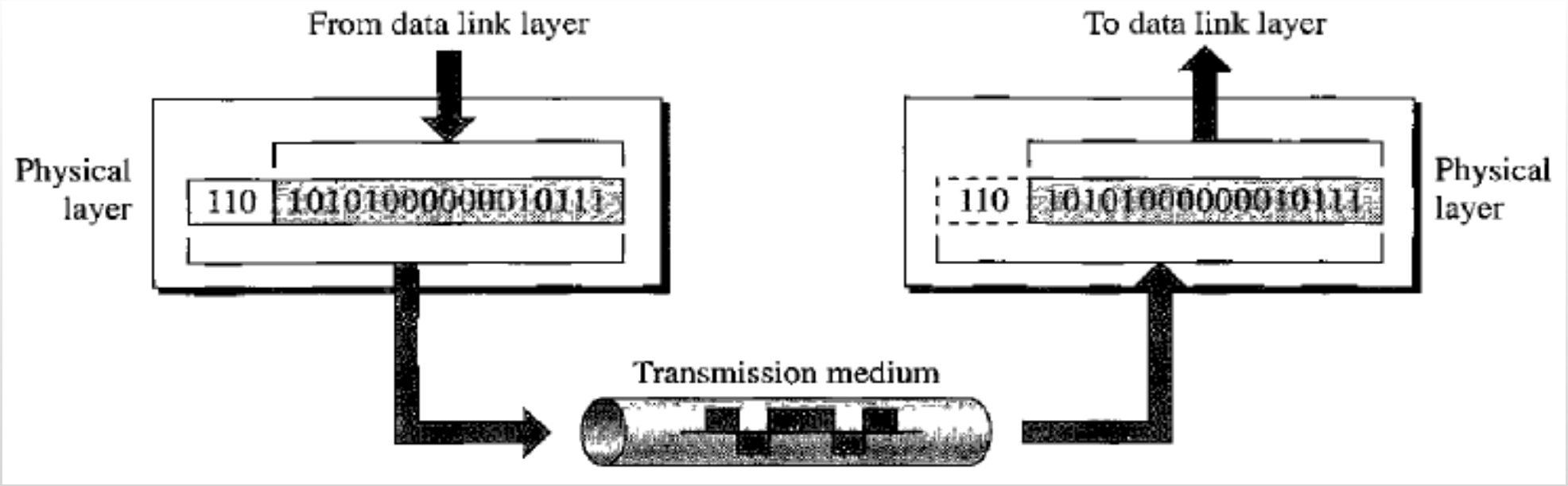
1. **Physical Layer**: This layer is responsible for sending and receiving bits of data between devices.
2. **Data Link Layer**: This layer makes sure that data is sent correctly from one device to another.
3. **Network Layer**: This layer is responsible for routing data packets between networks.
4. **Transport Layer**: This layer makes sure that data is sent correctly from one application to another.
5. **Session Layer**: This layer sets up, manages, and ends connections between applications.
6. **Presentation Layer**: This layer makes sure that data is in the right format for the application to understand.
7. **Application Layer**: This layer is where the user interacts with the network.



The OSI Model Function: Physical Layer

Physical Layer:

- Responsible for movements of individual bits from one hop (node) to the next.
- Representation of bits:-
  - ✓ data consists of stream of bits (0's & 1's),
  - ✓ Bits must be encoded into signals-electrical or Optical
  - ✓ Define type of encoding- how 0s and 1s are changed to signals.





The OSI Model Function: Physical Layer

Physical Layer:

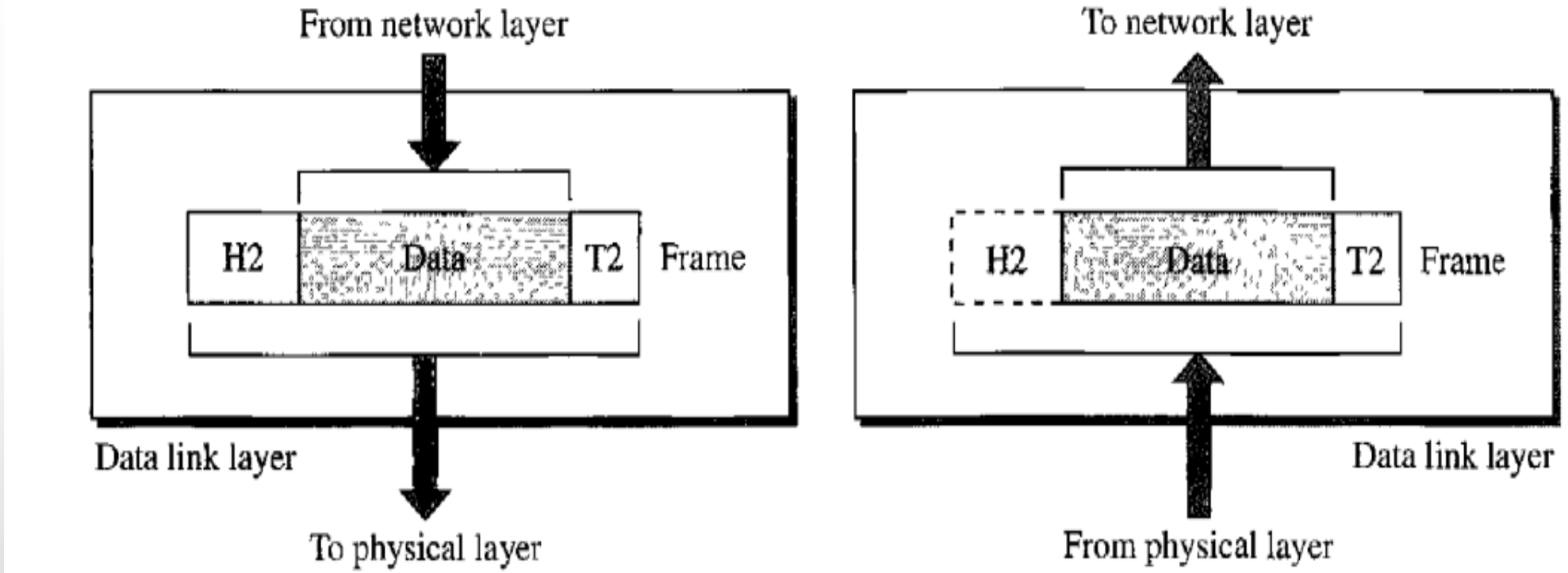
- Data rate (transmission rate):
  - ✓ the number of bits transfer each second.
- Synchronization of bits:
  - ✓ Not only same bit rates but clocks must also be synchronized.
- Line configurations:
  - ✓ Point-to-point configuration,
  - ✓ Multipoint configuration.
- Physical topology.
- Transmission mode.



The OSI Model Function: Data Link Layer

Data Link Layer:

- Responsible for moving frames from one hop (node) to the next.



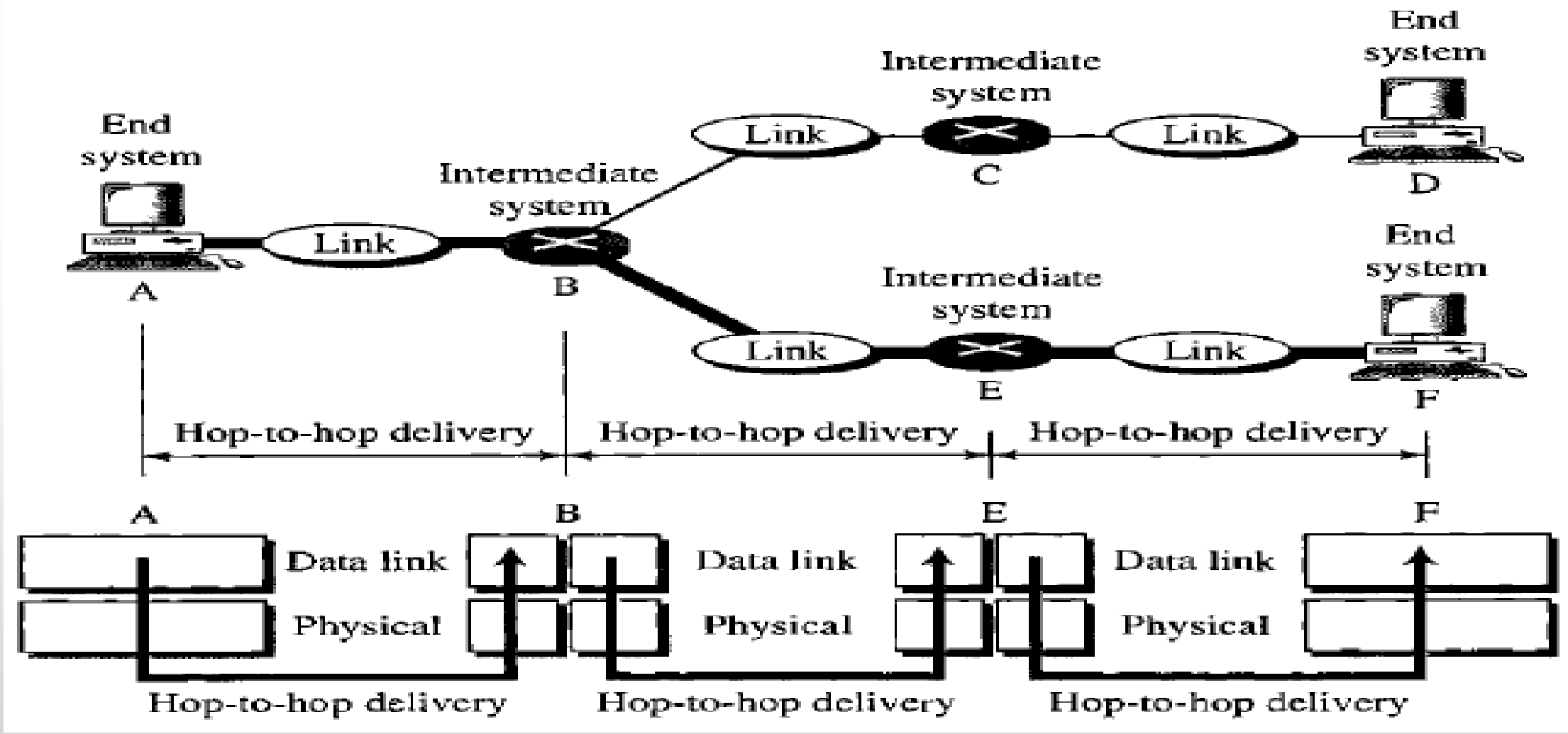


The OSI Model Function: Data Link Layer

- **Data Link Layer:**
  - Other Responsibilities includes:
    - ✓ Framing:
      - Breaks the stream of bits into manageable data units called frame.
    - ✓ Physical Addressing:
      - If frames are distributed into different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame.
    - ✓ Flow Control:
      - Prevents fast sender from overwhelming a slower receiver.
    - ✓ Error Control:
      - Returns positive / negative acknowledgement to the sender.
      - It adds reliability to physical layer by adding mechanism to detect and retransmit damaged or lost frames.
      - It also prevent duplication of frames.

The OSI Model Function: Data Link Layer

❖ Hop-to-hop delivery by Data Link Layer:

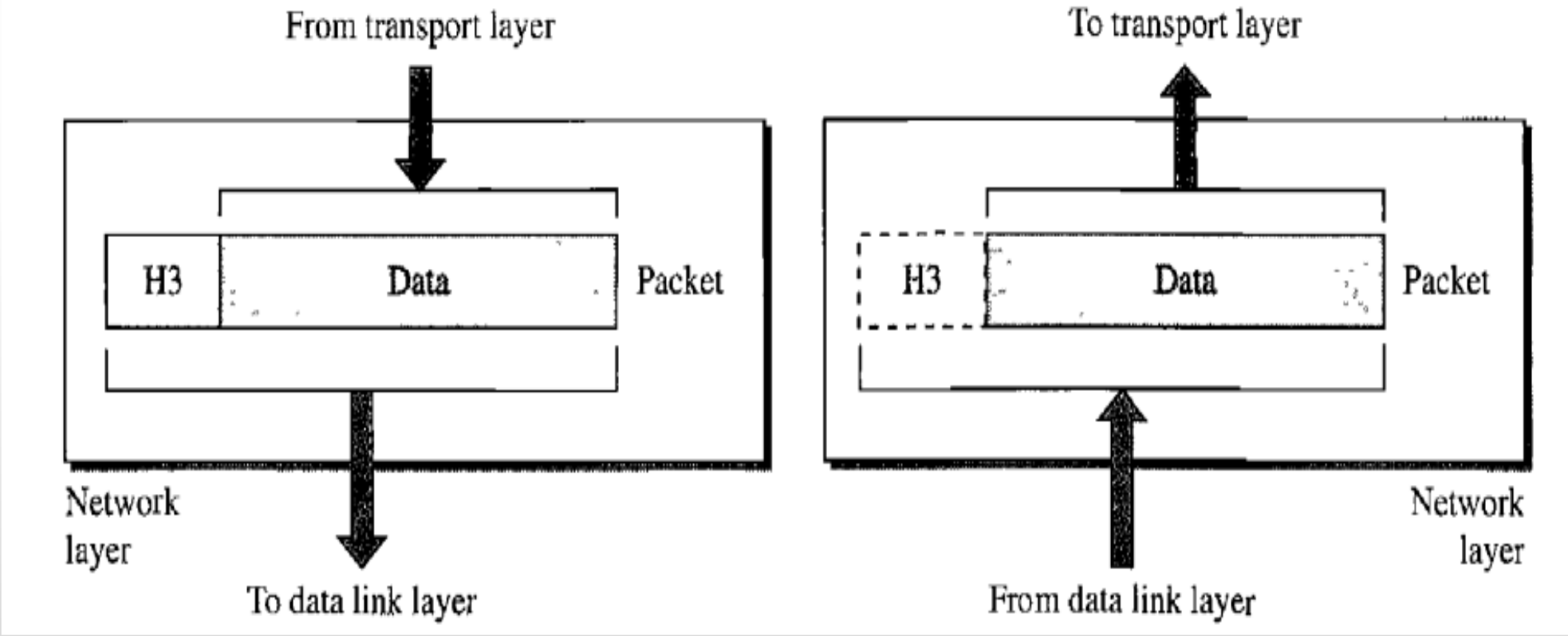




The OSI Model Function: Network Layer

❖ Network Layer:

- ➢ Responsible for the ***source-to-destination delivery*** of packet





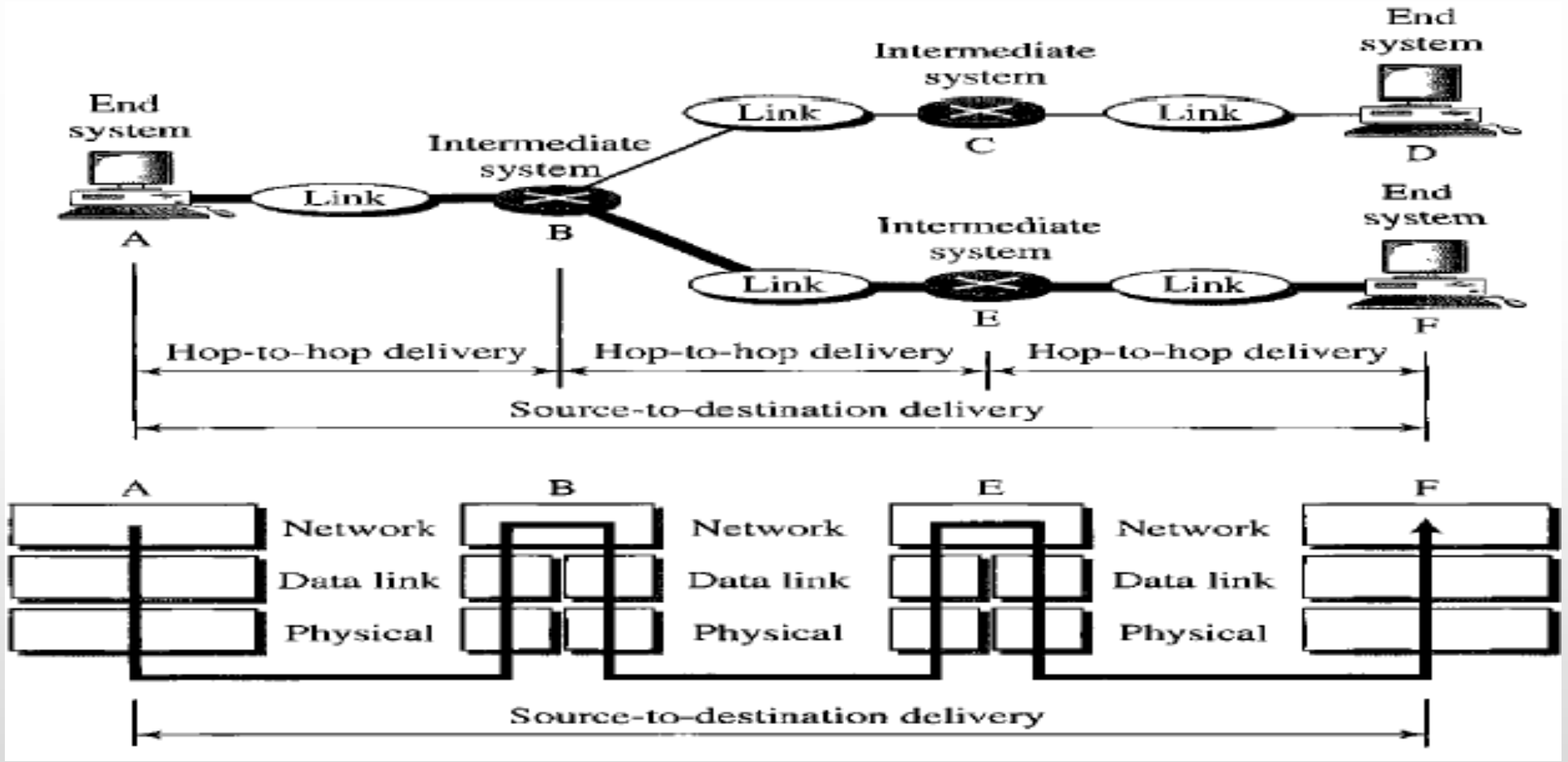
The OSI Model Function: Network Layer

- **Network Layer:**
  - Other responsibilities are:
    - **Logical addressing/Network addressing**
      - ✓ Physical addressing implemented by data link layer handles addressing problem locally.
      - ✓ If packet passes the network boundary, we need **Logical address** to distinguish the source and destination system.
    - **Routing**
      - ✓ In internetworks (a network of networks), a connecting device called routers/gateways/switches route the packets to their final destination.
      - ✓ Network layer provides routing mechanism.



The OSI Model Function: Network Layer

- **Source-to-Destination delivery of packet by Network Layer:**

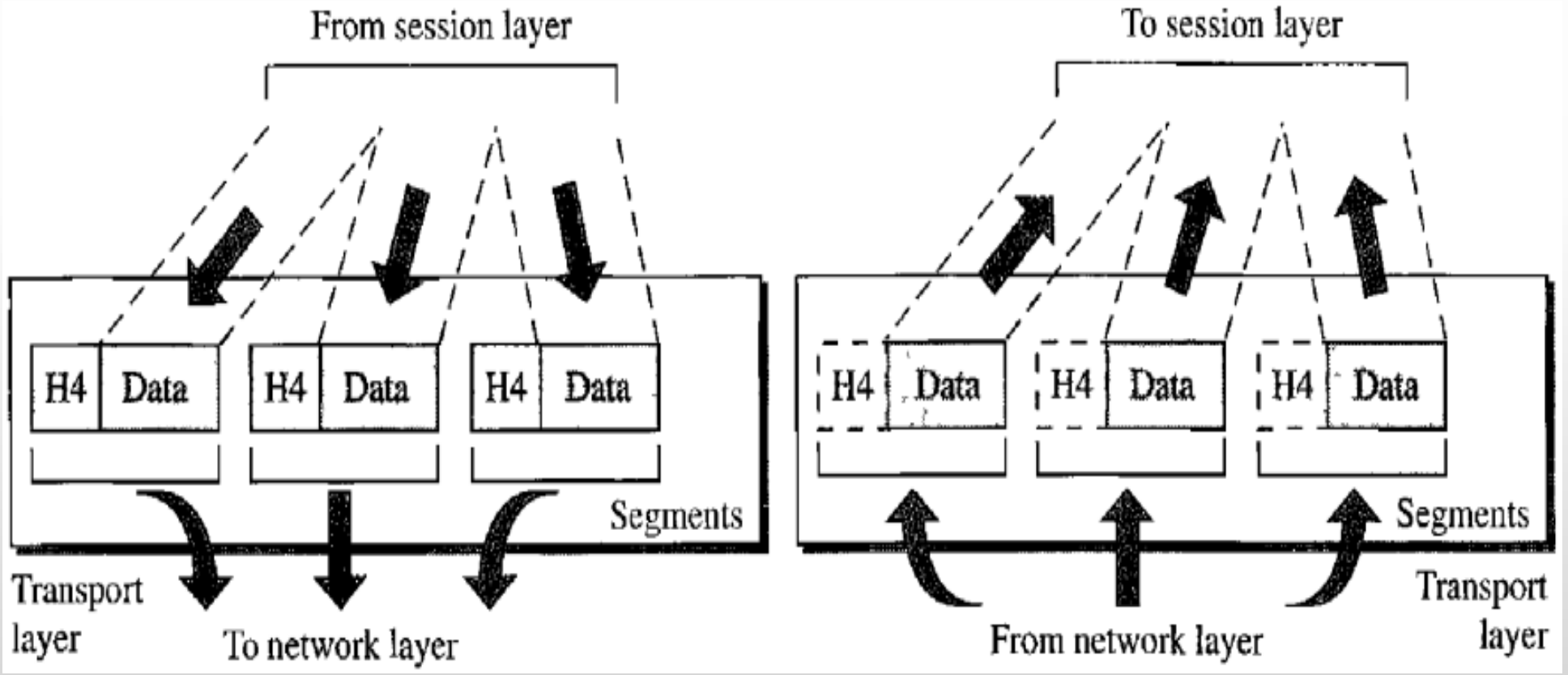




The OSI Model Function: Transport Layer

❖ Transport Layer:

- ➢ Responsible for the ***source-to-destination*** ***/process-to-process/*** ***end-to-end*** ***delivery*** of the entire message.





The OSI Model Function: Transport Layer

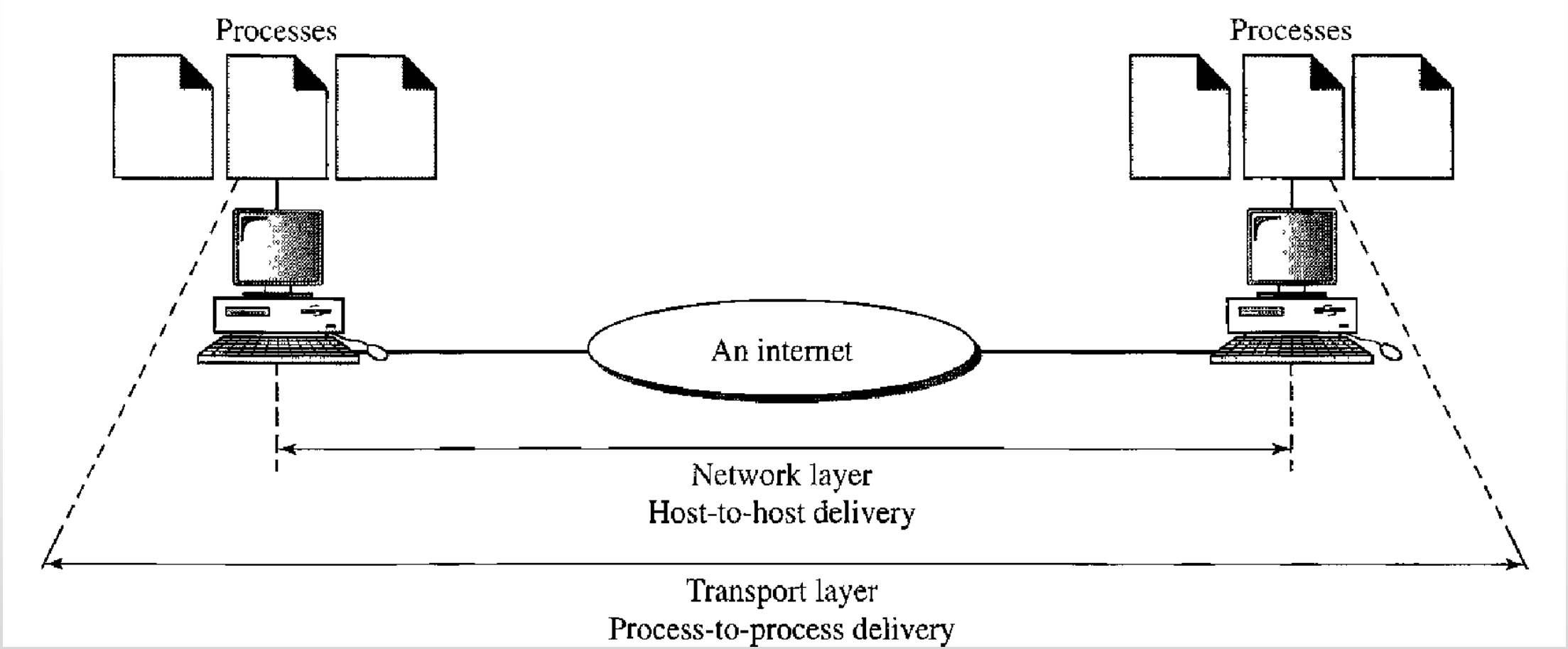
Transport Layer:

- Other responsibilities are:
  - **Service-port addressing:** port address to deliver a message.
  - **Segmentation and reassembly**
    - ✓ A message is divided into transmittable segment with each segment containing sequence number for reassembly the segmented message at the destination machine.
  - **Connection control:** either connectionless or connection oriented.
  - **Flow control and error control.**



The OSI Model Function: Transport Layer

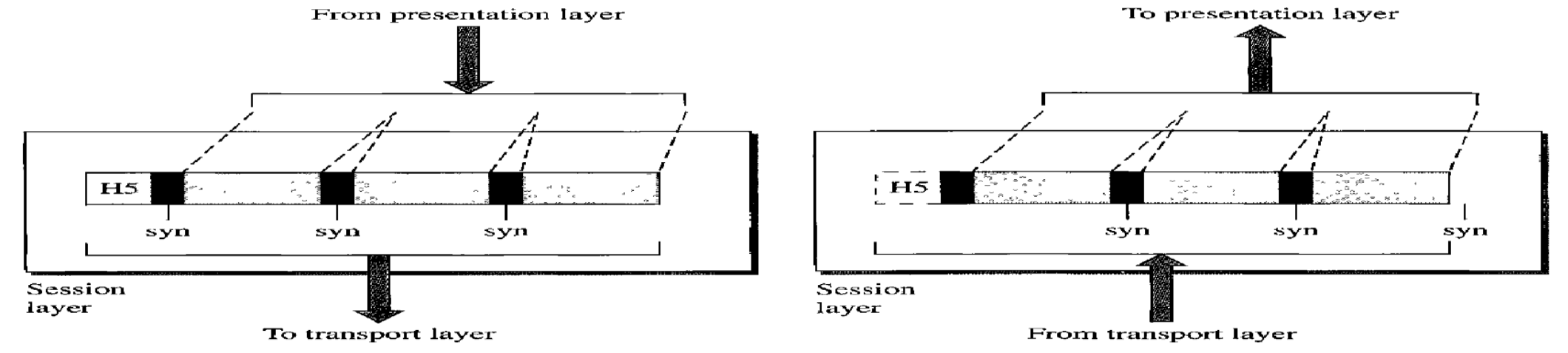
- **Process-to-process delivery message by the transport Layer:**





The OSI Model Function: Session Layer

- **Session Layer:** network dialog controller/ Interhost Communication.
  - Responsible for dialog control and synchronization (Establishes, manages and terminate sessions between applications).
    - ✓ **Dialog Control**:
      - It allows two systems to enter into dialog.
      - The communication between two processes take place either in half-duplex or full-duplex.
    - ✓ **Synchronization**:
      - Add check point (synchronization point) to the streams of data.





The OSI Model Function: Presentation Layer

Presentation Layer:

- Responsible for providing services to the user and data translator.
- **Specific responsibilities are:**
  - ✓ Character code translator:- ASCII to EBCDIC
  - ✓ Data Conversion:- bit order, integer to floating point etc.
  - ✓ Data compression:- reduces the number of bits that need to be transmitted on the network
  - ✓ Data encryption:- encrypt data for security purposes→ password encryption



The OSI Model Function: Presentation Layer

- **Application Layer:**
  - Responsible for providing window for the user and application processes to access network services.
  - **Some of the commonly used functions:**
    - ✓ Resource sharing and device redirection
    - ✓ Remote printer access
    - ✓ Network Management
    - ✓ Directory Services
    - ✓ Electronic Messaging



Recapitulation

We have learned:

Layer #	Name	Encapsulation Unit	Devices/Components	Description
7	Application	Data	PC	Network processes to application
6	Presentation	Data		Data Representation
5	Session	Data		Inter-host Communication
4	Transport	segments		End-to-end Connections
3	Network	packets	Router	Host-to-host delivery
2	Data Link	Frames	Bridge, switch, NIC	Access to Media
1	Physical	bits	Repeater, hub, transceiver	Binary Transmission



References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan



See you in next class…